

ME 333 - MECHANICS OF MATERIALS
PRACTICE PROBLEMS 3 - STRESS-STRAIN RELATIONS - SEP 16, 2025

Practice Problems:

From Prescribed Text - 3.1, 3.4, 3.6, 3.7, 3.9, 3.11, 3.15, 3.18.

- (1) Prove that the principal axes of stress and strain tensor coincide for an isotropic linear elastic material.

- (2) A unit cube of material is subjected to a principal stresses in three dimensions. Find all the conditions under which these stresses will not produce a volume change. Assume the material to be isotropic and linearly elastic.

[Answer] $\nu = 0.5$ or $(\sigma_1 + \sigma_2 + \sigma_3) = 0$

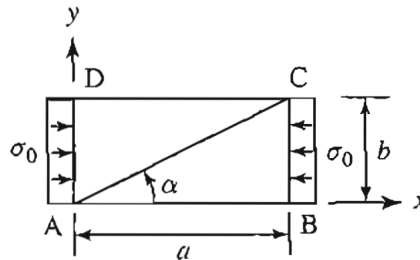
- (3) Derive the relation for Bulk modulus K as a function of :

(a) Young's modulus E and Poisson ratio ν

(b) Lamé's constants λ and μ

[Answer] (a) $K = \frac{E}{3(1 - 2\nu)}$, (b) $K = \frac{(3\lambda + 2\mu)}{3}$

- (4) A linear elastic plate with modulus of elasticity E and Poisson's ratio ν is subjected to a uniform compressive stress σ_0 as shown in Figure 4, such that at all points the only non-zero stress is $\sigma_x = \sigma_0$. Then: a) find the change in slope (Δ) of line AC as a function of a , b , ν , E and σ_0 , b) what is Δ if the material is incompressible?



[Answer] (a) $\Delta = \frac{b}{a} \left[\frac{(1 + \nu)\sigma_0}{(E - \sigma_0)} \right]$, (b) $\Delta = \frac{3b}{2a} \left[\frac{\sigma_0}{(E - \sigma_0)} \right]$

- (5) Draw a graph to show how the mean normal stress $p = (\sigma_1 + \sigma_2)/2$ varies with the shear stress $\tau = (\sigma_1 - \sigma_2)/2$ in a material which yields according to the Mises Yield criterion.

[Answer] The graph is an ellipse in $p - \tau$ space.

- (6) The state of stress at a point in the material is given as:

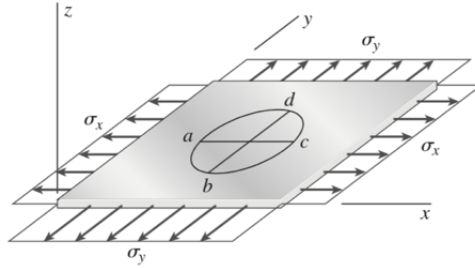
$$\begin{bmatrix} \sigma_{xx} & 0 & 0 \\ 0 & 3.5 & 0 \\ 0 & 0 & 5.5 \end{bmatrix} \times 10^8 \text{ Pa}$$

If the yield stress under uniaxial tension is 7×10^8 Pa, what minimum value should σ_{xx} be for the material to yield according to: a) Mises condition, and b) Tresca condition.

[Answer] (a) -2.28×10^2 MPa, (b) -1.5×10^2 MPa

- (7) A circle of diameter d is etched on a square brass plate of side l and thickness t , as shown in Figure. Forces are applied to the plate, producing uniformly distributed normal stresses $\sigma_{xx} = 2\sigma$

and $\sigma_{yy} = \sigma$. Calculate the following quantities: (a) change in diameter Δac that is along x , (b) change in diameter Δbd that is along y , (c) change in thickness Δt , and (d) change in volume ΔV . Assume E as the Young's Modulus and ν as the Poisson ratio.



[Answer] (a) $\frac{(2 - \nu)}{E} \sigma d$, (b) $\frac{(1 - 2\nu)}{E} \sigma d$, (c) $-\frac{3\nu}{E} \sigma t$, (d) $\frac{3(1 - 2\nu)}{E} \sigma l^2 t$